

Data Analytics (and Data Driven Decision)

Descriptive and Predictive Analytics



UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA



DISIM
Dipartimento di Ingegneria
e Scienze dell'Informazione
e Matematica

Andrea Manno
andrea.manno@univaq.it

Department of Information Engineering, Computer Science and Mathematics,
University of L'Aquila

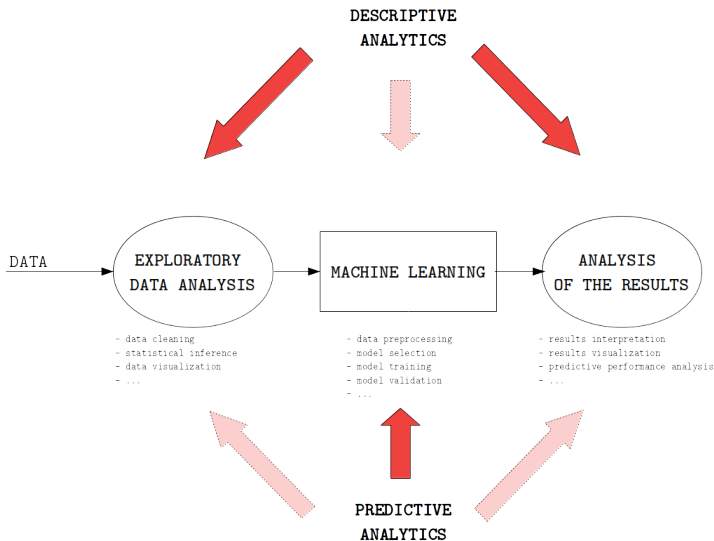


Overview

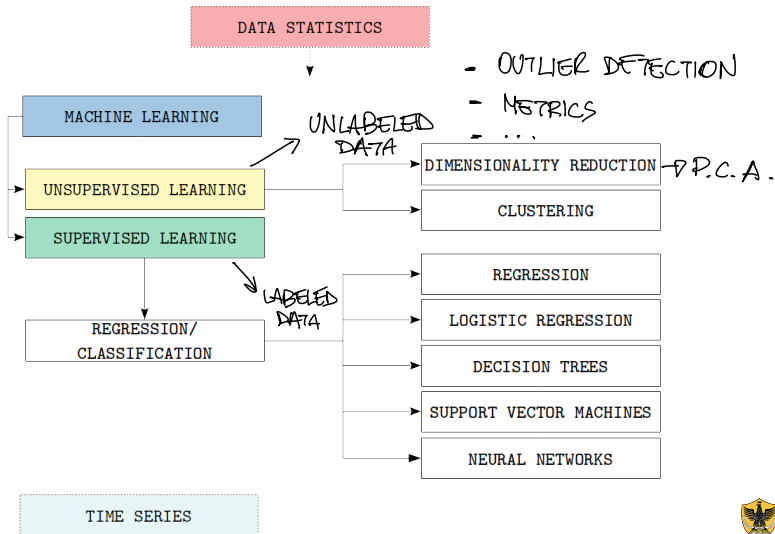
- 1 Introduction
- 2 Basic statistics
- 3 Multivariate statistics and Principal Component Analysis
- 4 Unsupervised learning: Clustering
- 5 Supervised learning
- 6 Overview on Time Series

Data Analytics: Descriptive and Predictive Analytics

a typical data driven decision process ...



Course in a glance



Machine Learning is about learning from **data** (predictive models)

- data collected, manipulated and cleaned
- good machine learning algorithms are used
- not enough...
 - ▶ data are **partial** (sampled) and **noisy** (errors)
 - ▶ big data: properly sampled? unbiased?

statistics needed with data!

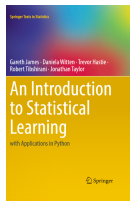
introductory part of the course on **basic statistics**

General course information

- material used during lectures (slides, exercises) is self-contained (sufficient for the exam preparation)
- reference books

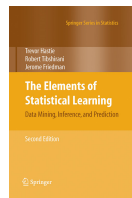
'An Introduction to Statistical Learning'

www.statlearning.com



'The Elements of Statistical Learning'

www.web.stanford.edu/~hastie/ElemStatLearn



DEMOS

- (annotated) slides and exercises published on the Teams course channel
- exercises are Python notebooks (can be replicated with any software/tool you prefer (Matlab, Excel, R, ...))
- **office hours** by appointment to be arranged by mail:
monday 11:00-13:00, friday 11:00-13:00

Exam information

Data Analytics determines 50% of the final mark

- **written exam** (up to 30 points):
 - ▶ multiple choice questions test
 - ▶ open-ended questions test
- **mandatory group project** (up to 1 point):
implementation and presentation of a Data Analytics work on a dataset provided by the lecturers

Overview

- 1 Introduction
- 2 Basic statistics
- 3 Multivariate statistics and Principal Component Analysis
- 4 Unsupervised learning: Clustering
- 5 Supervised learning
- 6 Overview on Time Series

Basic statistics concepts

arguments of this part

- DISTRIBUTIONS: SPREAD AND SIZE
- RANDOM VARIABLES
- SAMPLING
- INFERENCE
- RESAMPLING

Distributions: spread and size

how a **variable** is distributed in a **population**

- POPULATION OF INDIVIDUALS
- VARIABLE

WEIGHT
OF THE
PORTION
OF RICE



statistic is about synthesis, how to synthesize distributions?

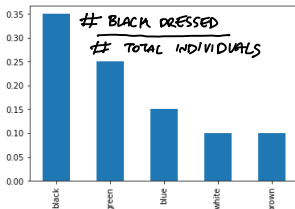
different representations according to

- type of variables: continuous, categorical, ordered,...
- variable values, frequencies,...
- ...

DISTRIBUTION
OF VALUES

idx	value
1	6.86
2	0.62
3	4.53
4	9.95
5	8.88
6	9.58
7	3.62
8	7.06
9	5.29
10	4.65
11	9.16
12	1.32
13	4.97
14	5.72
15	4.47
16	8.56
17	3.58
18	9.69
19	6.87
20	9.31

DISTRIBUTION
OF FREQUENCIES



DISTRIBUTION OF
FREQUENCIES FOR CONTINUOUS VAR.



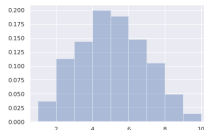
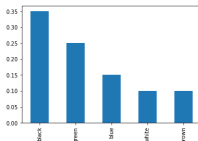
INDIVIDUALS, UNITS, OCCURRENCES, ...

a distribution is a way of representing how a variable is distributed in a population



which measures are important in a distribution?

idx	value
1	6.86
2	0.62
3	4.53
4	9.95
5	8.88
6	9.58
7	3.62
8	7.06
9	5.29
10	4.65
11	9.16
12	1.32
13	4.97
14	5.72
15	4.47
16	8.56
17	3.58
18	9.69
19	6.87
20	9.31



spread (variability): how much the values of the variable vary in the population?

given a population of n individuals (units, occurrences, events, ...) with (observed) values $x_1, x_2, x_3, \dots, x_n$, how to measure variability?

- total variability $\sum_{i=1, n} \sum_{j=2, n, j > i} |x_i - x_j|$
 - average variability $\frac{1}{\frac{n(n-1)}{2}} \sum_{i=1} \sum_{j=2, n, j > i} |x_i - x_j|$
- spread based on **differences among values** $\rightarrow \binom{n}{2}$



another way to measure spread: **distances with a reference value 'a'**

A DIFFERENT MEASURE FOR EACH d VALUE

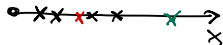
$$\underbrace{\sqrt[d]{\frac{1}{n} \sum_{i=1,n} |x_i - a|^d}}_{\text{DISTANCE}} \quad \text{or} \quad \frac{1}{n} \sum_{i=1,n} |x_i - a|^d \quad (1)$$

- (1) grows with the differences between values x_i and a
- (1) represents an overall distance measure between the values and the reference

once chosen a distance how to determine the reference value (**center**) w.r.t. that distance?

↘ d VALUE

- center is something intermediate between values...
- center is not too far from any of the values...



$$c_d = \arg \min_{a \in \mathbb{R}} \sqrt[d]{\frac{1}{n} \sum_{i=1,n} |x_i - a|^d} \quad \text{or} \quad \arg \min_{a \in \mathbb{R}} \frac{1}{n} \sum_{i=1,n} |x_i - a|^d \quad (2)$$

if we replace each value x_i with c_d the loss of information is minimized!

each distance has its center: the **size** (centrum, dimension) of the distribution

$d=2$ (EUCLIDEAN)
 • $\arg \min_{a \in \mathbb{R}} \frac{1}{n} \sum_{i=1,n} |x_i - a|^2$

MEAN

$$\mu = \frac{1}{n} \sum_{i=1,n} x_i$$

CONTINUOUS
VARIABLES

$d=1$ (MANHATTAN)
 • $\arg \min_{a \in \mathbb{R}} \frac{1}{n} \sum_{i=1,n} |x_i - a|^1$

MEDIAN

$$\xleftarrow{50\%} \bullet \xrightarrow{50\%}$$

ORDERED
VARIABLES

• x x x x x

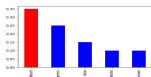


CATEGORICAL
VARIABLES

$d=0$

• $\arg \min_{a \in \mathbb{R}} \frac{1}{n} \sum_{i=1,n} |x_i - a|^0$

MODE



$$\begin{aligned} & \swarrow x_i \neq a \Rightarrow |x_i - a| \neq 0 \Rightarrow |x_i - a|^0 = 1 \\ & \searrow x_i = a \Rightarrow |x_i - a| = 0 \Rightarrow 0^0 \approx 0 \end{aligned}$$

spread + size = good synthesis of the distribution!

summing up...

POPULATION OF n INDIVIDUALS
WITH VALUES $x_i, i=1, \dots, n$ (continuous, categorical, ...)

CHOOSE THE DISTANCE
(SPREAD MEASURE)

$$\begin{aligned} &|x_i - a|^2 \\ &|x_i - a|^1 \\ &|x_i - a|^0 \end{aligned}$$

COMPUTE THE SIZE

MEAN
MEDIAN
MODE

EXAMPLE: 2 different populations...

idx	age
1	35
2	36
3	34
4	35
5	33
6	37
7	36
8	31
9	38
10	35

Figure: population 1: mean=35

idx	age
1	68
2	70
3	65
4	71
5	67
6	1
7	2
8	2
9	3
10	1

Figure: population 2: mean=35

the spread values are 2 and 35.0365!